

Standardizing the Requirements Management Process

ISD 503 Project Report

Submitted to:

Dr. Robert Bordley, Faculty Advisor
Integrated Systems and Design

Prepared by:

William Frith, Mr.
ISD Graduate Student
227 Hilltop Ridge Dr.
Madison, AL 35756

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Executive Summary

Warfare in flight is not solely the ability to place munitions on target. The US Navy understands the need for a multi-prong approach to air warfare and has developed aircraft to carry out various mission tasks. The EA-18G Growler is deployed to degrade and destroy an enemy combatant's ability to see potential threats through radar jamming capability. However, the ability to utilize current radar jamming capabilities in-flight requires that the radar jamming capability be powered independently of the aircraft due to the large power demand required for operation. This power requirement is satisfied through an independent air powered ram-air turbine power generation source.

The air powered ram-air turbine power generation unit is a flight product – i.e. those hardware and software items that are placed onto systems that fly – and requires rigorous evaluation, tracking and validation of requirements. Inadequate requirements development - especially in the domain of failure modes and flight safety - and lack of management of requirements can lead to overages in project cost, missed delivery dates, products that do not fulfill customer needs. More importantly these errors can produce failures resulting in loss of system, loss of life, and inability to fulfill mission.

The systems engineering effort on this project will incorporate a model-based systems engineering approach for requirements development, traceability and validation. This will help to alleviate issues normally observed in traditional document-based systems engineering approaches. Additionally, failure modes and effects analysis will occur within the systems model ensuring that all potential failures are adequately identified, mitigated and flowed to design of the unit.

1.0 Introduction and Current State

This is one of the largest and most scrutinized projects our company has taken on. While our company, CFD Research, has a large amount of experience in various fields of scientific research and traditional fields of engineering, systems engineering is a field that is new and untouched within our business. As such, it is imperative that this first formal dive into systems engineering set the stage and expectation for all other projects the company takes on. It is also pertinent to note that 4 full-time employees were planned to be employed on the execution of systems engineering tasks. However, hiring has been slow and largely unfruitful, leaving us with a team of 1.5 full time employees. I make up 1 of the 1.5 employees.

This project aims to fully define the methodology of requirements management that will be leveraged for design, manufacturing, and qualifying the Air Power Generation (APG) unit. This methodology will similarly be employed on future programs requiring systems engineering. This area was chosen for further evaluation since it is perhaps the most pervasive use of MBSE within industry. Requirements management is also a critical area of focus for systems engineering during this stage of the project.

1.1 Overview of the Project

As of September 13th, 2024 we are contractually obligated to design and build the APG, air power generation unit, for L3 Harris's Low Frequency Radar Jamming pod that will be employed on the US Navy's EA-18G Growler aircraft. The APG will sit at the front of the pod and be responsible for generating useable power for the radar jamming unit for a given flight envelope. We are required to deliver the first working prototype within 9 months of contract award, with the remaining 19 prototypes delivered by the end of 48 months. This compressed timeline provides a number of challenges both in the technical and programmatic solution.

To date, a total of approximately 300+ requirements have been flowed down initially from the prime contractor – L3 Harris – to our company for the APG unit. These requirements were analyzed for technical risks and whether they follow our current design. Additionally, no functional architecture was flowed from the prime contractor, so the employment of reverse engineering was needed to build a functional architecture. We were able to derive 5 system level functional requirements for the APG unit, 15 sub-system functional requirements, and numerous element and sub-element functional requirements.

Prior work in developing the RFP and SOW, as well as having built a single prototype mid-band APG unit, has provided a well-informed physical architecture for the system. This architecture was further refined by grouping the components into appropriate sub-systems. These sub-systems – and components – have been related to our derived functional requirements utilizing the 'Satisfy Matrix' within Cameo.

1.2 Review of Current State Design / Literature Review

CFD Research has previously built an APG unit for the Mid-Band Frequency Radar Jamming unit. This prior experience has allowed the company to develop a strong conceptual design for the physical architecture and functional requirements, as well as understanding the highest technical risks. However, the scope on the previous project was only for a single prototype production unit that only had to withstand ground testing. This single unit prototype design and production removed the need for quality requirements needed for serial production; and because it was limited to ground test, did not carry additional safety and reliability requirements needed to satisfy air worthiness testing. It is paramount that systems engineering be more closely tied into system definition to ensure all requirements in these areas are flowed to design engineering, manufacturing and test engineering.

Additionally, literature on standardized work processes was reviewed to help highlight the potential benefits gained. Standardization of work processes is needed to facilitate efficient, safe work methods and eliminate wastes, while maintaining quality (Kasul and Motwani, 1997). This statement alone indicates that increases in work efficiency, the elimination of wasteful work and consistent quality output are the benefits reaped when employing standardized work processes. Many companies would benefit from the application of standardized work; however, there is an initial cost in the way of man-hours needed to establish these processes and then to maintain these processes. These cons are often hyper-inflated and deter many from employing standard work. While we are resource limited at CFD Research, we also understand that the buy-in and maintenance costs will be paid off within the first few months of this project as we continue to bring on new personnel and expand the scope of our systems engineering tasking.

1.3 Stakeholders

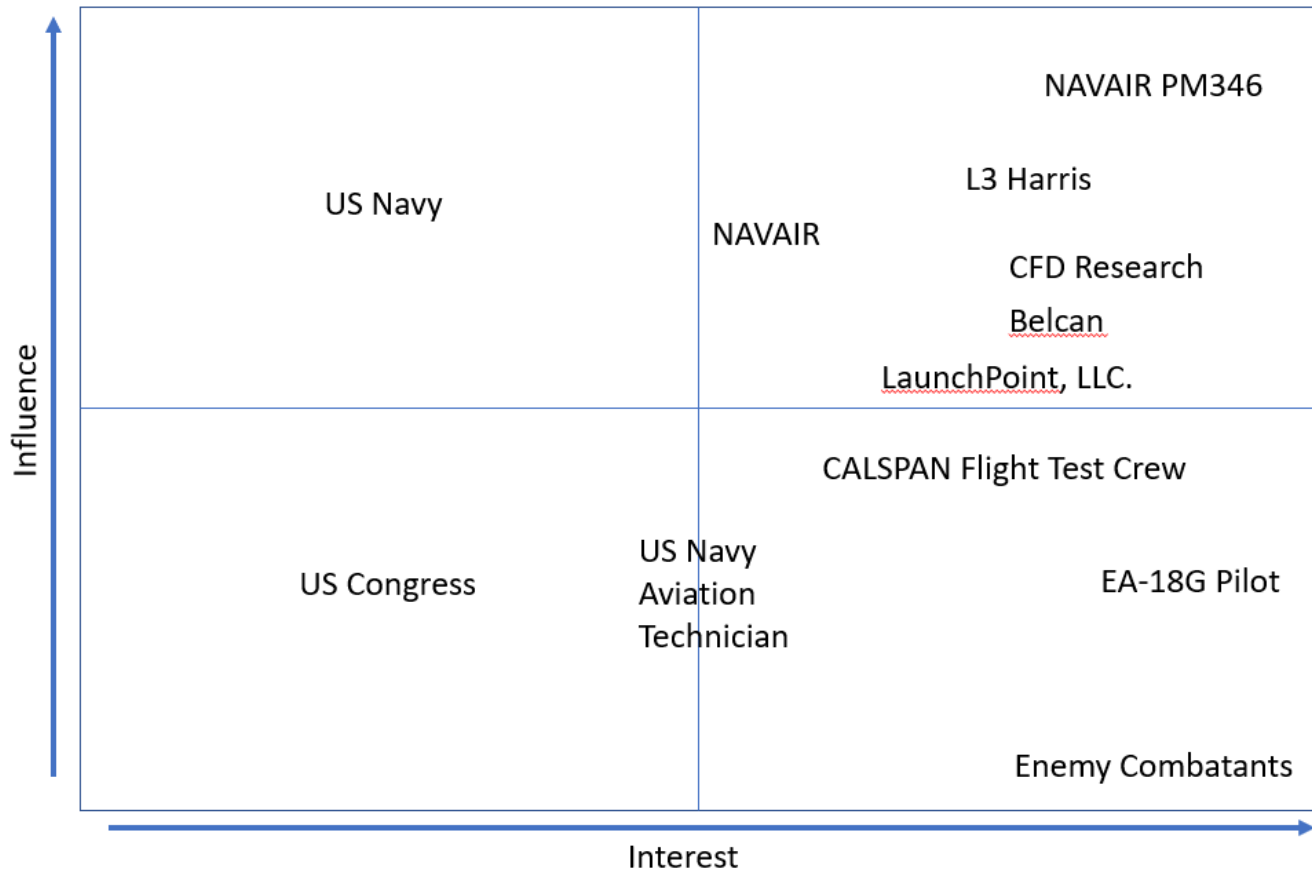


Figure 1: Stakeholder Matrix

1.4 Scope

The scope for the purposes of this project shall be limited to requirements management. Requirements management in this sense shall involve gathering, documenting, analyzing, managing change, tracing, and validating and verifying all requirements. The requirements management process begins in the design ideation phase of the project and continues throughout the lifecycle of the project and/or product.

The project will utilize MBSE and its inclusion will be integral to our requirements management process. Certain data connections or routine processes of uploading data from templated spreadsheets shall be utilized where data connections cannot be made or are impractical for this project.

1.5 Project Deliverables

The goal of this project is to define a standardized process for requirements management at CFD Research that can be leveraged on future projects. This will include a “lessons learned”

section that should be utilized to refine the requirements management process. While innocuous to most large companies, this is of great benefit for a company that has never formally addressed systems engineering and has no standard work process in place for requirements management. The establishment of a standard process not only allows us to leverage the process to other projects, but additionally allows us to establish a best practice, or baseline, that can then be iterated and improved upon.

1.6 Project Timing

This paper occurs at the beginning of a four year, \$50 million dollar project. During this four-year period a total of 20 prototype APG units will be delivered to L3 Harris, and subsequently the US Navy for testing and evaluation. During the time of this paper being written, Project Kickoff was completed along with the following deliverables from our systems engineering team:

- Compiled list of requirements that have not been incorporated into design (compliance)
- Complied list of requirements that have high technical risk
- Physical system architecture incorporated in the systems model
- Functional system architecture incorporated in the systems model
- Requirements imported into systems model
- Relational matrices created to connect model domains to one another
- Draft Integrated Test Plan
- Draft VCRM (Verification Cross Reference Matrix)
- Draft Safety Assessment Report
- Draft Safety Data File
- Draft FMECA (Failure Modes and Effects Analysis
- Mechanical Interface Control Drawing
- Software Interface Control Document
- Draft sparing list based on depot level availability

The above deliverables were completed by a team of four persons, including myself, in which I served and continue to serve as the lead for systems engineering activities.

2.0 Assumptions and Methodology

In this project we assume the following:

- CONOPs developed by NAVAIR and PMA-234 were utilized by L3 Harris in the development of technical specification for the NGJ-LB-APG (Next Generation Jammer Low Band Air Powered Generator pod)
- L3 Harris technical specification is flowed down to us after iterative feedback and acceptance by NAVAIR and PMA-234
- Technical specification provided to CFD Research is utilized to flow down initial requirements for sub-contractors working on APG unit. The sub-contracted portions of the APG unit only involve one of seven sub-systems, which is the generator, or power generation unit.
- Technical specification flowed to CFD Research was not evaluated for failure modes and effects analysis. This requires that FMECA be performed and that there will likely be derived requirements from the FMECA to support robust and reliable operation.
- Technical specification flowed to CFD Research was preliminarily evaluated for system hazard analysis. This hazard analysis did cover a finite set of failures that are relegated to the APG unit.

Additionally, we assume that all other sub-systems that make-up the NGJ-LB-APG pod outside of the APG unit are designed, manufactured, tested and managed by L3 Harris.

The methodology applied for requirements management will be best evaluated from the definition utilized for requirements management: defining, gathering, validating, and controlling all requirements.

2.1 Requirements Gathering

Requirements gathering shall involve direct interaction with the next higher level in project hierarchy, i.e. sub-contractor should engage the prime contractor or prime contractor should engage with the customer. The “direct customer” should provide, at a minimum, a scope of work document and specification document. If a technical specification is not provided, then a subsequent process shall need to be developed to document how to perform requirements definition.

Additionally, those persons gathering requirements should determine if any analysis work was done to inform the technical requirements that were flowed down, and if so what was the scope of said analysis.

2.2 Requirements Documenting

Catia No Magic (Cameo) shall be utilized as the single source of truth for all requirements. All requirements flowed from the “direct customer” to CFD Research shall be imported or manually entered into a requirements table within Cameo. Additionally, it should be noted that all requirements shall have a source from where they are derived set as a parameter within the model. The source shall either be embedded within the system model, or a hyperlink shall be provided, to allow access to the source.

All known future sources of requirements, largely derived requirements, shall be identified and incorporated into the systems architecture within the model. As the design matures, these placeholders shall be fulfilled through either a Cameo plug-in or through data connection to an external source (i.e. excel spreadsheet mapped to some import within Cameo).

Functional architecture shall be generated within the system model. Any conceptual design solution that exists shall be evaluated for decomposition into sub-systems and components. This conceptual design architecture shall be generated within the model.

2.3 Requirements Analysis

Requirements analysis requires the establishment of relationships between architectures and the determination of compliance with design and technical feasibility. There are three major domains that we need to establish relationships between:

- 1) Requirements
- 2) Design Solution (or physical hardware)
- 3) Functional (or operational domain)

To establish our Design Compliance matrix, the design solution architecture shall be mapped to requirements utilizing a satisfy matrix within Cameo. This matrix shall utilize a relationship of directly satisfy, indirectly satisfy and not satisfied. For purposes of this satisfy matrix, indirectly satisfy should be utilized when it is possible that the design solution fulfills the requirement but:

- clarity is needed in the requirement definition
- requirement is in conflict with another requirement
- referenced document in requirement definition has not been provided

Requirements not mapped to the design solution shall be flagged as non-compliant. A register of all non-compliant (not satisfied and indirectly satisfied) requirements shall be generated. Any design solution sub-system or component not mapped to any requirement should additionally be added to the register as requiring evaluation of removal from the design solution. All entries with a designation of “indirectly satisfied” within the non-compliance register shall be put forward for resolution from the “direct customer”. For those entries with a designation of “not satisfied”, a mitigation plan shall be established, actions generated with dates for completion

within the mitigation plan, and assigned owners for each action. Recurring meetings to address status of non-compliant requirements shall be established at some interval until all non-compliant requirements are satisfied by the design solution.

Requirements are also analyzed from a technical risk perspective. To make as informed a decision on the technical feasibility of the requirements, a quorum was established to evaluate and define path forward. Our quorum consisted of integrated product team leaders within air vehicle engineering, software and controls, manufacturing and quality, program management and systems engineering, integration, and test. Additionally, the chief engineer on the project attended and provided guidance during the evaluation. All requirements were categorized by high, medium and low designations for technical risk. Requirements designated as high risk were added to an additional register and required to have a burn down plan (plan to minimize risk) logged and assigned owners. Requirements designated as low risk required no further action.

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Additionally, we shall establish relationships between the functional domain and the design solution. Depending on the level of the functional architecture sub-systems or components that demonstrate “not satisfied” by any function are indicative of being either unnecessary or interfaces between components and/or sub-systems.

2.4 Management of Change

For requirements requiring either deletion, addition, or modification, a change request process shall be utilized. This change request shall require the initiator to define which requirement is being addressed, what is the suggestion (deletion, addition, or modification), and to present this information to a configuration control board which meets once every two weeks or can be requested ad hoc if immediate resolution is required. All changes shall be recorded in a log which defines the requirement(s) changed, the date on which the configuration control board approved the change, and the scope of the change (modified, deleted, added, etc...).

2.5 Requirements Traceability

Traceability of requirements is managed through Catia No Magic. All initial requirements flowed to CFD Research from L3 Harris were imported from Excel into Catia No Magic. All derived requirements shall be generated through Catia No Magic and managed through Catia No Magic. All other documents are considered artifacts of and do not reflect the source of truth for requirements, with the exception of changes generated by the Navy or L3 that are then flowed to us.

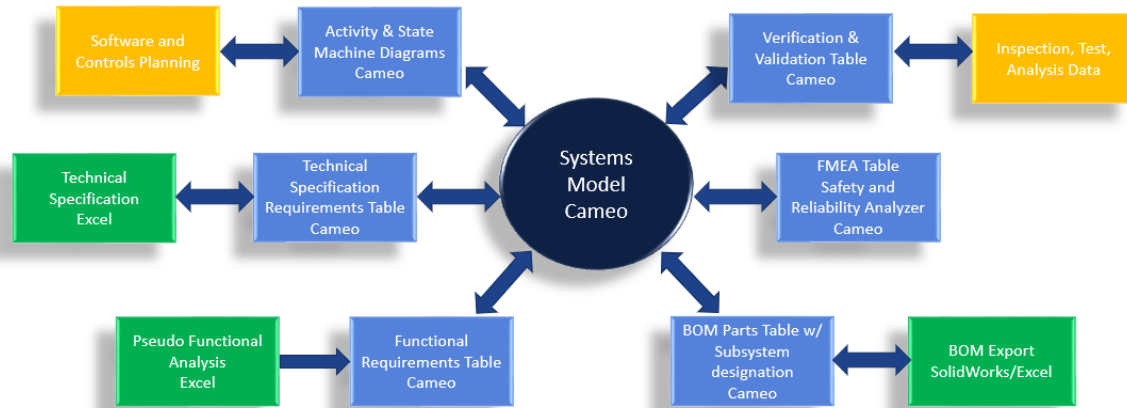


Figure 2: System Model Data Flow Diagram for NGJ-LB-APG

2.6 Verification and Validation

Requirements verification and validation in the context of requirements management, and for the purpose of this paper, shall not include the actual verification and validation of requirements. However, it shall include the generation and alignment of three separate documents:

- Verification Cross Reference Matrix (VCRM)
- Integrated Test Plan (ITP)
- Integrated Master Schedule (IMS)

The VCRM provides a designation of the verification and validation methodology (inspection, analysis, demonstration, test, NHA) that must be employed for each individual requirement. It should be noted that the designation “NHA” means that at our level of verification and validation that there exists some preliminary toll-gate, but that verification and validation of that requirement will only be resolved at the next higher level of assembly. For our APG unit, this would mean potential inspection, analysis, demonstration or test occurs at the NGJ-LB-Pod level.

The VCRM was provided within the technical specification and was imported into Catia No Magic from Excel. The VCRM exists as a requirements matrix within the model. The integrated test plan aligns requirements and/or groups of requirements to which test(s) (vibration, salt fog, built-in test, etc...) will be required, the equipment required and any applicable standards for testing of said requirement(s). This exercise is beneficial as it helps to minimize the number of tests that need to be performed since a single test could be utilized for verification and validation of multiple requirements. Additionally, the integrated test plan will help to identify which tests are destructive versus non-destructive. This is important to ensure time and costs are minimized since we are aligning execution of tests on prototype units prior to subjecting the prototype units to destructive testing.

As stated at the beginning of the paper, there are 20 prototype units required to be built and delivered. These prototype units have varying levels of design completeness associated with

them and therefore have varying levels at which they can be used to verify and validate requirements. Our integrated test plan shall therefore align tests, which align to requirements and/or groups of requirements, to prototype units.

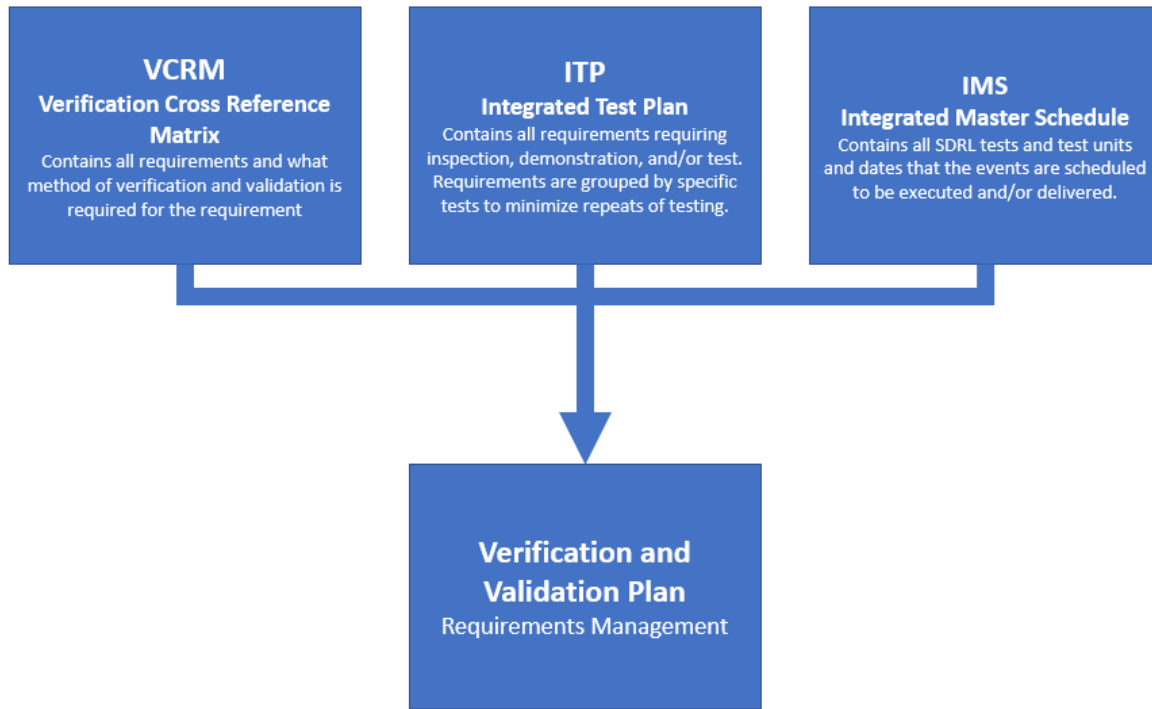


Figure 3: Verification and Validation Data Integration Diagram

3.0 Results and Discussion

As section two was presented in a segmented format to address the large scope of requirements management, section three shall similarly be constructed. This section shall provide results largely in a discussion format as most findings will be adjustments to a process.

3.1 Requirements Gathering

In future endeavors, it would be of great benefit to receive any and all pertinent information from the prime contractor and/or customer. The requirements gathering methodology stated in section 2.1 would have been sufficient; however, due to lack of resources and manpower on the L3 Harris side of this project, getting additional key documents outside of the technical specification and the scope of work was almost impossible. File sharing was also a cumbersome and overly long process, which meant limited ability to even share the documents once the documents requested were found. While this did slow progress, having a better list of documents to look for outside of the technical specification and scope of work, would have been helpful to expedite receiving needed information.

During the process of gathering requirements, it was unclear as to whether the requirements had a rationale assigned to them. A number of requirements seemed to suggest a particular design solution, but this was not explicitly stated in any supporting documentation. Examples include the need for backup power to ensure that certain physical components would actuate in the event of a loss of power to the APG unit. While backup power is one possible solution to deal with this failure mode, the reality is that a mechanical solution, such as a spring-loaded device, could place the device that needs to be actuated in a safe state without the application of electrical power.

No CONOPs was provided which would likely have played a further role in understanding why certain built-in test functions were desired. To add, lessons learned from the US Navy's mid-band pod, have yet to be shared though they do exist. Having a technical resource within our customer's organizational structure would help tremendously in tracking down additional supporting information.

3.2 Requirements Documentation and Analysis

Requirements documentation and analysis was carried out as defined in section 2. For the most part the requirements analysis was successful in what was stated as our goal in requirements analysis. Functional architecture was almost immediately generated upon receiving the technical specification, and physical architecture was generated within the model utilizing the bill of materials for the request for proposal computer aided design file. Relationships between functional architecture and physical architecture were established through a satisfy matrix, and all requirements were mapped to a physical sub-system allowing us linkage between all three domains. Additionally, all requirements were evaluated for design compliance and technical risk.

- Compliance
 - Requirement is fulfilled with current design (green)
 - Requirement is included but needs recovery plan to pull into compliance (yellow)
 - Requirement is not included in current design (red)

- Note: Table is only evaluating the major physical sub-system that requirements are mapped to. Many requirements are allocated to multiple physical sub-systems.

Compliance Score Card		
91	100	172
Compliance Rating ▾		Count of Requirements
⊕ Green		91
⊖ Red		172
All		9
APG Controller		146
Battery		1
Cosmetics		3
Exhaust Doors		5
Support Structure		5
Turbine		3
⊕ Yellow		100
All		51
APG Controller		6
Cosmetics		2
Exhaust Doors		3
Generator		2
Support Structure		23
Turbine		13
Grand Total		363

Figure 4: Requirements Compliance Score Card

- Technical Risk/Feasibility
 - High - Needs burn down plan
 - Medium - Needs further analysis or clarity
 - Low - No concerns
- Ranking generated based on:
 - Able to include requirement
 - Able to verify and validate requirement
 - Clarity required for requirement
 - Conflicting requirement
 - Documents referenced in requirement missing

Note: Table is only evaluating the major physical sub-system that requirements are mapped to. Many requirements are allocated to multiple physical sub-systems.

Risk Score Card		
High	Medium	Low
3	29	331
Risk Ranking	▼	Count of Requirements
High		3
All		1
APG Controller		1
Turbine		1
Medium		84
All		10
APG Controller		46
Battery		1
Exhaust Doors		4
Generator		1
Support Structure		12
Turbine		10
Low		276
Grand Total		363

Figure 5: Requirements Technical Risk Score Card

One item that needs to be addressed in future work is how best to handle analysis of software requirements, particularly states and modes and built-in testing. It is difficult to understand the interplay of switching from one state to another state and understanding what criteria are required for changing of states when these items are represented as lines of text in a document. There needs to be a preliminary step to generate flow diagrams to represent these change in states with their associated inputs and outputs. Additionally, built-in testing should be represented in some graphical format to easily digest what functions and what sensors will likely be required to be included in design.

For the most part, success in requirements analysis was largely the result of brute force. The deliverable timeline drove the team to have to participate in lengthy review sessions to determine

design compliance and technical risk that each requirement presented. The time applied to large group reviews could be minimized through a number of improvements.

- 1) Apply an initial functional (logical) architecture to all requirements
- 2) Apply an physical architecture and associate all requirements to piece parts or sub-systems depending on design maturity
- 3) Require independent reviews by IPT lead with the functional block architecture and/or physical architecture of the system

By applying the above steps, independent reviews can happen as an initial review. When the quorum is called to review the design compliance and technical risk, the review should happen much quicker and resolve issues of finding time slots to accommodate all IPT leads.

3.4 Management of Change

There are no real findings from the management of change process largely because we did not have to leverage our MOC process for requirements. Due to the nature of our contract, L3 would routinely meet with our systems engineering team where we would discuss issues with requirement verbiage, clarity, or scope creep. These issues were kept in a log, along with the rationale for the change and L3 would then update the technical specification which would flow down to us in the form of a newly revised technical specification.

As we develop our own technical specification for a number of the APG unit sub-systems, the MOC process developed and state in section 2 will be enacted whenever issues arrive from our internally derived requirements.

3.5 Requirements Traceability

Requirements were imported into the model as set forth in section 2. While not solely contained with the topic of requirements traceability, adding of additional parameters to requirements within Catia No Magic is a bit of a cumbersome task. This is important to note because parameters for requirements may contain the logical linkage between other domains or may serve as triggers for parametric evaluation programmed within the model. In a more abstract sense, those parameters may be place holders that would be utilized much further in the product lifecycle, i.e. reliability centered maintenance activities like sparing. This inability to flow changes like addition of a parameter to all requirements, does mean that work often occurs in spreadsheets or other documents outside of the model. It then becomes imperative in designing a process that ensures that the information contained within these documents finds it way into the model. As it currently stands, our systems architect is being tasked with figuring out how to import those changes. It is easy to see how such a work process would inevitably invite errors in either setting of parameters for requirements, or adding in new derived requirements to spreadsheets but not importing those into the model. Training on the usage of Catia No Magic and the establishment of a defined process to incorporate changes that occur outside of the model are both needed items to ensure traceability of requirements, and their various relationships to other domains, is maintained.

3.6 Verification and Validation

The process outlined in section 2.6 has proven to be the most difficult to achieve success in and is still ongoing. While there was some initial success in aligning the VCRM to the ITP, many of the definitions of what the 20 different prototype units are, are poorly defined. This has led to massive back and forth with L3 and the Navy to determine what exactly a structural prototype is versus a captive mass model prototype versus a jettison mass model prototype versus a qualification unit versus an operational prototype unit. Remember that our alignment was to align requirements states as requiring test, demonstration or inspection to a specific test or set of tests within the ITP. Then we were to align the tests within the ITP to deliverable units required per the scope of work and dated as needing delivery within the IMS. As stated, it is very difficult to align testing and then verification and validation activities for requirements when the thing to serve as the medium for testing is poorly defined.

It would be fair to say that requirements gathering in this instance was not fully executed and all parties carry some amount of responsibility for this. In retrospect, at Project Kickoff a larger discussion should have occurred to address uncertainties about the definition of prototype units (i.e. what is the rationale for having this test unit). The compressed deadlines for deliverables coupled with minimal staffing - as hiring was occurring - meant that only those things required by contract to be reviewed and evaluated were. MBSE could likely help to some degree if the information was presented within the model as gaps and then disseminated to the team. Evaluating root cause and how to improve would involve more fully investigating the definition of prototype units to be utilized for testing would be paramount and ensuring that personnel are hired prior to, or that some means of managing workload through hiring of temporary workers (i.e. other subcontractors) is established before contract award. However, all of this should be clarified that the integrated test plan and therefore it being enacted is not a deliverable until FDR (final design review) which is not until August 2025.

4.0 Recommendations

The largest recommendation is to create documentation, including many visual aids such as flow diagrams, to support a base level process for requirements management. Without an established baseline to work against it is difficult to generate specific changes that can add additional benefit from a cost and time perspective. Establishment of baseline documents for the systems engineering process would address, and likely resolve, many of the issues found through the various exercises required in requirements management.

Baseline documentation would then be subjected to peer review within the company to solicit feedback on improvement. These documents would be subjected to revision control, published to all CFD employees, and trained on by those required to execute these tasks.

As observed in this paper, each of these individual focus areas of requirements management presented unique problems that have been recorded and can now be included in our initial revision of requirements management documentation. Additionally, it is apparent that more training needs to occur with our systems engineering team to understand how to better utilize the MBSE tool.

5.0 Impact/Financial Benefits

The impact of documenting this process has a broad and cascading value. As we are a company that has never taken on systems engineering work within our contracts, it is imperative that we define the standard by which we will perform systems engineering. Some of these items may seem mundane, but they have far reaching impacts if not defined before starting of a project. Standardization in work processes has known value in many industries and disciplines. Systems engineering is no different.

While difficult to quantify, standardization of these processes has been of great value in discussions with L3 Harris – our prime contractor – and with the Navy. By having a written and approved process in place, our customers have a greater sense of our team being able to complete the tasks and that the quality of our work will be accurate and consistent. Additionally, it provides a space in which our customer can provide commentary on the effectiveness of our process and generate insight into potential improvements.

Additionally, as we are beginning this process, we are hiring personnel that may have limited background within systems engineering, which provides an even greater need for standardization in our work processes. As a new hire, before starting work on a project, the employee will need to become familiarized with the documents that govern how we intend to execute systems engineering in all its various stages in the process or product lifecycle. This will remove ambiguity from tasks assigned to new employees and better align individual success with project success. Additionally, new employees may realize gaps in the process that need to be changed or improved upon, allowing almost immediate success for the new employee and a greater degree of competency to be established in our documentation.

For those currently working, the benefits are similar in nature. The how of executing systems engineering tasks are well defined and bounded, allowing personnel to more efficiently allocate their time and energy. Baseline documentation can then be revised as gaps or new, more efficient means of executing said tasks are discovered.

6.0 Summary/Conclusions

In summary, requirements management is multi-faceted and requires a great deal of data gathering, sorting, aligning of data sets, and understanding of rationale. It additionally requires that these things be done within a short time period, and with a fair degree of accuracy and it may require the use of personnel that are either new to the company or contracted out to provide support. These constraints coupled together speak to a need to ensure that a process is established and defined. The application of an established process that had been peer reviewed and subjected to real world deployment will provide existing and future employees with clear guidance on how to execute. Ambiguity and lack of clarity in direction could result in needless hours, days, and weeks being spent without arriving at the target objective.

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